

Kyungdong University Global Website & CMS

Complete project documentation for the public React website and the PHP/MySQL content management system that powers it.

Frontend: kdu-global-website (React + Vite)

Backend: kdu-backend (PHP + MySQL / XAMPP)

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1. Project Overview

The KDU Global digital platform consists of two tightly integrated applications:

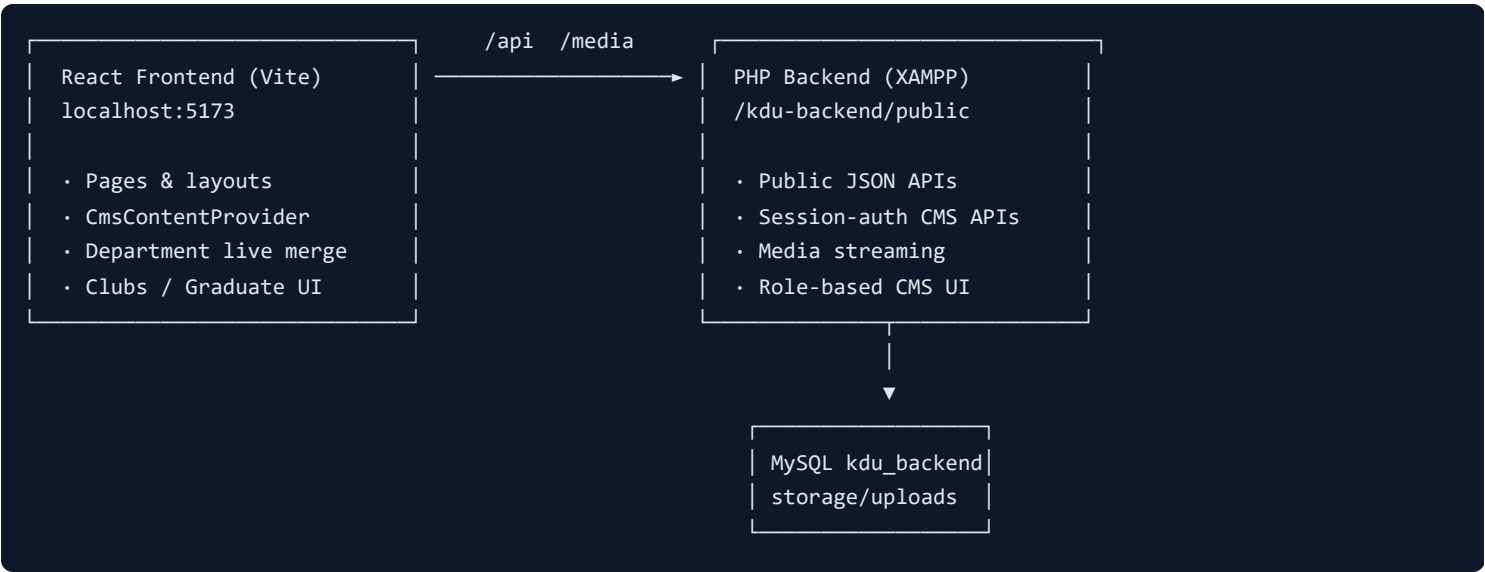
- 1. **Public website** — a modern React single-page application that presents academic programmes, graduate studies, engagement/clubs, research, admissions, and news.
- 2. **CMS backend** — a PHP/MySQL content management system used by Super Admins, Heads of Department (HODs), faculty members, Graduate School staff, International Office, and club administrators to manage live content.

Static fallback content in the React app ensures the site remains usable if the API is temporarily unavailable. When the CMS is connected, published content and faculty profiles override or merge with those fallbacks.

Goals

- One public brand experience for Kyungdong University Global.
- Department pages that HODs can update without developer intervention.
- Faculty profiles created once and listed under multiple programmes if needed.
- Club pages managed by club admins created only by Super Admin.
- Clear draft → publish workflow so unfinished edits never appear publicly.

2. System Architecture



Integration pattern

- In development, Vite proxies `/api` and `/media` to `http://localhost/kdu-backend/public`.
- Public pages use unauthenticated GET requests.
- CMS login uses PHP sessions with `credentials: 'include'`.
- Only **published** records are returned by public APIs.

3. Frontend Application

Path: `C:\Users\Mascud Mohamed\kdu-global-website`

3.1 Technology stack

Layer	Technology
UI library	React 19
Bundler / Dev server	Vite 6
Routing	React Router DOM 7
Styling	Custom CSS + Tailwind CSS 4
Motion	Framer Motion
Icons	lucide-react, react-icons

3.2 Directory layout

src/	
api/	Site, department, and clubs HTTP clients + mappers
components/	UI building blocks (department, graduate, engagement, layout...)
constants/	Navigation menus and shared links
context/	CmsContentContext (site-wide published content)
data/	Static fallback content (departments, graduate, clubs...)
hooks/	useDepartmentLiveData, useClubs, useFacultyProfile...
layouts/	Main, Department, Engagement, Research layouts
pages/	Route-level screens
routes/	router.jsx
styles/	Brand variables and component/page CSS

3.3 Main routes

URL	Purpose
/	Home (CMS-driven hero / headline / CTA)
/about , /admissions , /contact	Core institutional pages
/academics	Academics hub
/academics/undergraduate/:slug	Department hub + spokes
/academics/undergraduate/:slug/faculty	Department faculty directory
/academics/graduate	Graduate Courses (K-Global GSBTC)
/academics/graduate/faculty	Graduate faculty directory
/engagement , /engagement/clubs/:slug	Engagement & student clubs
/research , /research/:page	Research hub and spokes
/international-office	Student services / international office
/news-events , /sitemap	News hub and site map

3.4 Undergraduate departments

Six programmes are registered in the frontend data layer:

- Smart Computing
- International Business Administration (API code `iba`)
- English for Academic Purposes (API code `eap`)
- Artificial Intelligence
- International Hotel Management (API code `ihm`)
- Korean for Academic Purposes (API code `kap`)

Each department page merges static content with live CMS data (hero, curriculum, faculty, news, events, admissions, contact) via `useDepartmentLiveData`.

3.5 Graduate school

The graduate experience is modelled after the official KDU Global Graduate Courses page (K-Global Graduate School of Business and Technology Convergence). It includes hero, dean message, programmes, course outline, curriculum components, careers, faculty, scholarships, and admissions steps. Sub-navigation mirrors undergraduate department UX.

3.6 CMS content on the public site

- `CmsContentProvider` loads `GET /api/site/content` on app start and refreshes on window focus.
- Home uses CMS headline, tagline, intro, CTA, and hero image when present.
- About, Admissions, Contact, Navbar, and Footer consume CMS modules similarly.
- Media paths such as `/kdu-backend/public/media/...` are normalized to `/media/...` for the Vite proxy.

3.7 Brand & UI guidelines (in code)

- Primary blue `#0072BC`, accent red `#D62828`
- Typography tokens in `src/styles/variables.css`
- Department and graduate heroes use full-bleed imagery with overlay text
- Prefer merging CMS data over hardcoding when a field is published

4. Backend CMS & API

Path: `C:\xampp\htdocs\kdu-backend`

4.1 Technology stack

Layer	Technology
Runtime	PHP 8.2+ (plain PHP, no Laravel)
Database	MySQL / MariaDB (<code>kdu_backend</code>)
Auth	PHP sessions stored in MySQL
CMS UI	Static SPA under <code>public/cms/</code>
Entry point	<code>public/index.php</code>

4.2 Directory layout

```
kdu-backend/  
  config/          app, database, mail configuration  
  database/        schema migrations 001-012, install.sql  
  public/          API entry + CMS UI + media routes  
  scripts/         create_root, create_faculty_account, unlock_user...  
  src/  
    Auth/ Controllers/ Core/ Http/ Mail/ Rbac/ Services/  
    routes/api.php  routes/cms.php  
  storage/uploads/ faculty/ events/ news/ site/ clubs/
```

4.3 CMS portals

URL	Audience
/kdu-backend/public/cms/	Root, HOD, Dean, International, Club Admin
/kdu-backend/public/cms/faculty/	Faculty profile editors

4.4 Publish model

- **Draft** — editable in CMS, invisible on the public website
- **Published** — returned by public APIs; may set `published_at`
- **Archived** — retained but hidden from public listings
- Faculty profiles use `is_published` with an explicit Publish action

After publishing site content, refresh the React site (or wait for focus refresh) to see changes. The CMS success message reminds editors to open `http://localhost:5173`.

4.5 Media storage

Uploads are separated by purpose so faculty photos are never mixed with site heroes:

Purpose	Folder
<code>faculty_photo</code> , <code>faculty_cv</code>	<code>storage/uploads/faculty/</code>
<code>site_image</code> , <code>content_image</code>	<code>storage/uploads/site/</code>
<code>news_image</code>	<code>storage/uploads/news/</code>
<code>event_banner</code>	<code>storage/uploads/events/</code>
<code>club_image</code>	<code>storage/uploads/clubs/</code>

Public URL pattern: `/media/{folder}/{year}/{month}/{filename}`

4.6 Faculty multi-department affiliations

A faculty member creates **one** profile. Through `faculty_profile_affiliations` they can appear under multiple undergraduate departments and optionally under Graduate School.

- Home department remains on the user/profile for HOD ownership.
- Additional departments are selected in the Faculty Dashboard or when creating the account.
- Public lists: `/api/{department}/faculty` and `/api/graduate/faculty`.

4.7 Clubs CMS

- Only Super Admin / Root creates clubs (**Manage clubs**).
- Root then creates a `club_admin` user assigned to that club.
- Club admins edit only their club profile, news, and events, then publish.
- Frontend engagement club pages merge published API clubs with static fallbacks.

5. Database Schema

Database name: `kdu_backend`. Migrations live in `database/schema/`.

File	Contents
001	Departments, roles, permissions foundation
002	Users, sessions, password reset
003	Faculty profiles and related tables
004	Department events & news
005	Media files metadata
006	Audit logs
007	Early seed data
008	Enterprise CMS: clubs, content blocks, office scopes
009	Enterprise RBAC seed & content modules
010	Extra department CMS modules
011	Starter clubs seed
012	Faculty multi-department affiliations

Full install: `mysql -u root kdu_backend < database/install.sql`

6. Roles & Permissions

Role key	Who	Can manage
root	Super Admin	Everything: users, site content, clubs catalog, settings, audit
hod	Head of Department	Own department content, news, events, faculty accounts
faculty	Teaching staff	Own profile, photo/CV, publications, affiliations, publish
dean_graduate	Graduate School	Graduate content, events, news
international_office	International Office	International content, events, news
club_admin	Student club lead	Assigned club profile, news, events
*_event_news_manager	Legacy dept editors	Department events/news only

Permissions are enforced in `PermissionService` and `CmsGuard`. Users are scoped by `department_id`, `club_id`, or `office_scope`.

7. Operational Workflows

7.1 Super Admin publishes Home content

1. Sign in to CMS as Root.
2. Open **Website content** → Home Page.
3. Edit headline, tagline, intro, CTA, and upload a *site* hero image.
4. Click **Publish**.
5. Refresh the public site at `http://localhost:5173`.

7.2 HOD manages a department

1. Sign in as HOD (account linked to one department).
2. Edit department pages, news, and events; publish when ready.
3. Create faculty accounts under **Faculty accounts** (optionally tick extra departments / Graduate).
4. Faculty log into the Faculty Dashboard to complete and publish their profile.

7.3 Faculty multi-programme listing

1. Faculty opens Faculty Dashboard → Profile.
2. Under **Departments & programmes**, tick every programme where they teach.
3. Optionally tick **Also list under Graduate School**.
4. Save and Publish — one profile appears on each selected public faculty list.

7.4 Create a club and club admin

1. Root → **Manage clubs** → create club.
2. Root → **Users** → role Club Admin → assign club → temporary password.
3. Club admin signs in → My club page / news / events → Publish.

8. API Reference (Summary)

8.1 Public endpoints

Method	Path	Description
GET	/api/health	Health check
GET	/api/site/content	Published site content bundle
GET	/api/{dept}/content	Published department content modules
GET	/api/{dept}/faculty	Published faculty (incl. affiliations)
GET	/api/{dept}/news	Published department news
GET	/api/{dept}/events	Published department events
GET	/api/clubs	Published clubs
GET	/api/clubs/{slug}	Single published club
GET	/api/graduate/faculty	Faculty affiliated with Graduate
GET	/api/graduate/events	Graduate office events
GET	/media/...	Stream uploaded media

8.2 Auth & CMS (authenticated)

Area	Examples
Auth	/api/auth/login , /me , /logout , password reset
Site content	/api/cms/content (draft/publish)
Admin users	/api/cms/admin/users
HOD faculty	/api/cms/{dept}/faculty-users
Faculty self	/api/cms/faculty/me (+ children, photo, publish)
Clubs	/api/cms/clubs , /api/cms/club/profile
Office	/api/cms/office/{graduate international}/...
Media	POST /api/cms/media/upload

9. Installation & Local Development

9.1 Prerequisites

- XAMPP (Apache + MySQL/MariaDB)
- Node.js 18+ (recommended 20/22)
- PHP 8.2+
- Git

9.2 Backend setup

1. Place the backend at `C:\xampp\htdocs\kdu-backend`.
2. Start Apache and MySQL in XAMPP Control Panel.
3. Create database `kdu_backend` if needed.
4. Run migrations:

```
mysql -u root kdu_backend < database/install.sql
```

5. Copy `.env.example` to `.env` and set DB credentials, CORS origin `http://localhost:5173`, and mail settings.
6. Create Root user:

```
php scripts/create_root.php admin@kduglobal.edu "YourStrongPass123!" "Super Admin"
```

7. Verify: open `http://localhost/kdu-backend/public/api/health`
8. CMS: `http://localhost/kdu-backend/public/cms/`

9.3 Frontend setup

```
cd "C:\Users\Mascud Mohamed\kdu-global-website"
npm install
npm run dev
```

Open `http://localhost:5173`. Leave `VITE_API_URL` empty in development so the Vite proxy handles API calls.

9.4 Useful npm scripts

Script	Purpose
<code>npm run dev</code>	Local development server
<code>npm run dev:host</code>	Expose on LAN / tunnels
<code>npm run build</code>	Production build to <code>dist/</code>
<code>npm run preview</code>	Preview production build

10. Deployment Notes

- Build frontend with `npm run build`; host `dist/` on a static web server or CDN.
- Set `VITE_API_URL` to the public backend origin before building.
- Point Apache/Nginx document root (or alias) at `kdu-backend/public`.
- Ensure `mod_rewrite` is enabled and `public/.htaccess` is respected.
- Make `storage/uploads` writable by the web server user.
- Configure HTTPS and restrict CMS access if exposed publicly.
- Align CORS allowed origins with the production frontend URL.

11. Troubleshooting

Symptom	Likely cause / fix
CMS shows Published but site unchanged	Hard-refresh frontend; confirm <code>/api/site/content</code> has non-empty published fields; ensure Home maps hero/intro.
Broken image in CMS preview	Use Upload (stores <code>/media/...</code>); avoid pasting full XAMPP browser paths; confirm file exists under <code>storage/uploads</code> .
Faculty photo appears as home hero	Do not reuse faculty media for site hero; site uploads go to <code>media/site/</code> .
MySQL ERROR 2002	Start MySQL in XAMPP.
Import path / Vite resolve errors	Check relative imports from <code>src/pages/</code> use <code>../</code> not <code>../../</code> .
Faculty not listed on second department	Ensure affiliations saved and profile published; check <code>faculty_profile_affiliations</code> .
Club admin cannot be created	Create the club first under Manage clubs, then assign <code>club_id</code> .

Security reminder: Never commit real `.env` credentials or production passwords. Rotate temporary passwords after first login. Keep Root accounts to a minimum.